Uzair Arif

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Recent Economics & Mathematics graduate at IBA Karachi with hands-on experience building LLM systems, time-series forecasting pipelines, and production RAG applications. Interned at FBR, NCAI, and NASTP. Looking for junior AI/ML roles.

Education

Institute of Business Administration — BS Economics and Mathematics, CGPA : 3.25 *June 2026*

Key Courses: Foundations of Data Science, Linear Algebra, Probability Theory, Optimization Techniques, Applied Econometrics, Statistical Machine Learning, Stochastic Processes, Game Theory

Skills

Languages & Libraries: Python, SQL, Scikit-learn, XGBoost, CatBoost, PyTorch, Prophet, Darts, ZeroMQ

LLM / RAG: LangChain, FAISS, BM25, Reciprocal Rank Fusion, Groq (Llama 3), SciBERT, Optuna, Streamlit

MLOps & Tools: MLflow, FastAPI, SQLite, Docker, Git, Pytest, Ruff, Mypy, QGIS, Kaggle

Concepts: Time-Series Forecasting, Anomaly Detection, RAG, Multi-Agent Simulation, Drift Monitoring

Professional Experience

National Center of Artificial Intelligence (NCAI) — Research Intern *Jul 2025 – Aug 2025*

- Conducted literature review on Multi-Agent RL, public goods games, and Ostrom's governance principles; drafted a full research proposal on emergent institutional design in cooperative agent systems.
- Designed core simulation components including democratic voting on punishment rules, weighted voting schemes, and IPPO agent architecture with reward functions for norm emergence.
- Work laid the foundational groundwork for a final-year project on LLM-agent governance dynamics, later supervised at IBA.

Federal Board of Revenue (FBR) — Revenue Forecasting Intern *Jun 2025 – Jul 2025*

- Built 5 daily Prophet models (Income Tax, Sales Tax, Federal Excise, Customs Duty, Gross Collection) with Pakistan-specific Gregorian and lunar holiday effects and monthly seasonality.
- Tuned all models via Optuna hyperparameter search and evaluated on held-out validation sets using MAE and MSE to handle highly volatile government revenue patterns.
- Automated a multi-horizon Excel reporting pipeline with daily, weekly, and monthly breakdowns; deployed an interactive Streamlit dashboard to surface mid-month surges and fiscal deadline spikes for the team.

Ali Khan Consultants — Junior Data Analyst *Sep 2024 – Dec 2024*

- Identified 32% (60,000 kg) forecast discrepancy in procurement records, driving order delays and supply chain disruptions.
- Diagnosed 15-day gap from process to sales order (above industry avg) and 4-day lag to dispatch, highlighting warehousing inefficiencies.
- Found 41% of monthly revenue occurs in last 5 days; final-day orders 3.5× above expected daily average.
- Analyzed revenue concentration: 25 materials generate 80% of revenue; 63% from Shades FA White variants.

National Aerospace Science & Technology Park (NASTP) — AI Intern *Jun 2024 – Aug 2024*

- Engineered real-time data streaming and visualization tools using ZeroMQ and Python, reducing publish-subscribe latency to milliseconds.
- Built CNNs, linear regression, and SVM models from scratch, improving predictive accuracy by 10%.
- Created an anomaly detection model for ECG time-series data, achieving a 95% detection rate for anomalies.

Projects

Multi-Agent LLM Simulation for Climate Governance

- Simulated 26 heterogeneous LLM agents in a climate cooperation game; found collection vastly exceeded disbursement (34.5B vs. 424M) and final wealth inequality persisted (Gini 0.645), exposing redistribution design flaws in the LDF framework.
- Implemented Theory of Mind audits and behavioural consistency scoring that fed reputation back into agent prompts, creating social pressure while preserving anonymity; agents voted on constitutional rule changes mid-simulation.
- Built concurrent LLM inference pipeline integrating reputation tracking, gossip mechanics, and democratic governance across 26 agents using Python.

NYC Taxi Demand Forecasting — Production MLOps System

- Built production MLOps pipeline for real-time NYC taxi demand forecasting: Optuna-tuned XGBoost achieving Val R^2 of 0.949, tracked via MLflow Model Registry.
- Engineered persistent SQLite feature store with TTL pruning and WAL-mode concurrency; served predictions via FastAPI with batch and single-zone endpoints.
- Implemented KS-test drift monitoring to detect statistically significant distributional shifts in live demand data; simulated streaming replay against live API for end-to-end validation.
- Enforced production code quality: 36/36 tests passing across unit, API, and integration suites; zero ruff lint and mypy type errors; fully containerised with Docker.

IBA RAG Chatbot

- Built a modular, production-structured hybrid RAG system combining FAISS semantic search and BM25 keyword retrieval fused via Reciprocal Rank Fusion across 1000+ HTML files and PDF documents.
- Engineered custom ingestion pipeline with recursive chunking to preserve context boundaries.
- Integrated Groq LPU inference backend to achieve sub-second output generation latency across institutional documents.